# IM0802 – Responsible Artificial Intelligence
Assignment 2: “Paper Title Goes Here”

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## Abstract

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## 1 Introduction 350

Israel, which allocates 8.8% of its GDP to defense [1], invests heavily in the development of autonomous weapon systems (AWS), with their Prime Minister announcing a $100 billion investment in homegrown arms production last December [2]. Israel’s most well-known AWS is the Iron Dome air defense system, which intercepts incoming projectiles in real time. Building on this foundation in artificial intelligence (AI)-assisted defense technology, Israel has expanded into offensive targeting applications such as “Habsora”.

Habsora ("The Gospel") is an Israeli Defense Forces (IDF) AI-targeting system developed primarily by Unit 8200 within the Military Intelligence Directorate. First deployed during the May 2021 Operation "Guardian of the Walls", it generates target recommendations using a multimodal intelligence pipeline that encompasses signal intelligence, telephone metadata, satellite and drone imagery, social graph analysis, and life-pattern modelling [3, 4]. The system is not a single model, but an ensemble of multiple specialised algorithms fused together [4]. During the 2023-2024 Gaza conflict, Habsora was reported to generate over one hundred target recommendations per day, which in prior years required approximately twelve months of human analysts [3, 5].

Habsora’s AI capability is the probabilistic classification and prioritisation of locations and individuals as military objectives. It assigns confidence scores derived from large-scale pattern matching across a shared intelligence pool, rather than positive identification in the traditional intelligence analysis [6]. Military commands nominally authorise each recommendation, yet their review window documented for each target was approximately twenty seconds [3].

This creates a stakeholder dynamic with both ethical and legal significance. It’s output influencing both IDF commands, who operate under high cognitive load and time pressure; Palestinian civilians in the target environment bearing the lethal consequences and the broader international community which depends on International Humanitarian Law (IHL) compliance for the legitimacy of armed conflict. Steen et al. [7] offer a framework for this tension. They propose a two-axis model of Meaningful Human Control (MHC) which distinguishes between high automation and genuine human control, which both require adequate information and real freedom of action. The central argument of this report is that Habsora structurally undermines the second condition, and that this gap carries cascading implications across the technical, sociotechnical, and governance layers identified by Verdiesen et al. [8] in their Comprehensive Human Oversight Framework (CHOF).

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## 2 Ethical Issue Identification 750

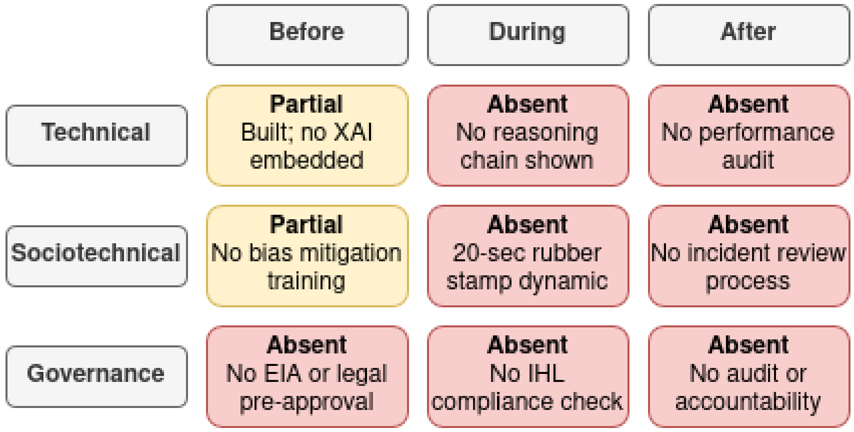


Figure 1: The CHOF framework for structuring ethical analysis across governance, socio-technical, and technical layers. Two yellow cells indicate where Habsora can be given partial compliance. The red cells indicate absence.

Verdiesen et al. [8]'s Comprehensive Human Oversight Framework (CHOF) was developed to assess accountability for autonomous systems whose internals are not directly inspectable. It divides oversight into a 3×3 matrix, before, during and after deployment across a technical, socio-technical and governance mapping. The resulting matrix surfaces gaps which remain hidden when 'human oversight' would be treated as a single requirement. Applying this framework to Habsora's implementation we get Figure 1.

### 2.1 Technical Layer

Habsora has been described as an ensemble of deep learning models processing multimodal data [3, 4]. Such architectures are inherently opaque, Arrieta et al. [9] characterise this as the accuracy/interpretability trade-off, where deep ensembles maximize predictive performance at the cost of transparency. While post-hoc explainability methods exist that could be applied to such systems, reports on Habsora's use indicate that no such methods are applied as the operators point of decision.

This opacity directly links to the legal consequences. Schmitt and Thurnher [6] establish that IHL's distinction principle requires positive identification of a target as being a military objective. In conjunction with the proportionality assessment being understood by the user. Without any interpretability the legal review is effectively empty.

The above failures map directly into the highest-ranked principles identified by Khan et al. [10]. Both **P-01** transparency and **P-03** accountability, further compounded by their challenges: lack of technical understanding among decision-makers **CH-06** and the absence of binding legal frameworks **CH09**. While this might not be a failure of intent, it is a structural property of opaque architectures. Combined with lacking operator-facing interfaces, this creates a system which can not provide the context for understanding.

### 2.2 Socio-technical

The Value Sensitive Design (VSD) framework addresses indirect stakeholders, those affected without directly using the system, such as civilians in the target environment. Palestinian civilians were not involved in its design, and their values were not elicited to ground the system's targeting logic. They remain non-consenting parties who face the consequences of the system's use without recourse. There is a large asymmetry in consequences between those operating the system and those subjected to it.

For IDF commanders, the conditions necessary for genuine human oversight are structurally not present. Steen et al. [7] argue that meaningful human control requires both sufficient information and meaningful freedom of action. Habsora undermines both: commanders receive recommendations without transparency into the underlying reasoning, and the short decision windows of the system leave almost no room for critical evaluation. Miller [11] shows that from social science, without understanding what went into a decision, the explainee cannot meaningfully interrogate it. Combined with the large amount of targets the system generates, this reduces what would normally require extended deliberation and proportionality assessments to a button press. Former IDF commanders have described this process as ‘rubber-stamping’ [12]. This confirms the presence of automation bias, which is when operators rely too much on AI outputs under time pressure [13].

The consequence is that IDF commanders formally bear moral and legal responsibility for strikes they cannot meaningfully evaluate, while the system’s opacity diffuses accountability in practice. Targeting decisions that once demanded careful human judgement are now effectively delegated to an algorithm. This prevents the human control which the IHL demands, and that Steen et al. [7] identify as a precondition for attributing responsibility.

### 2.3 Governance

Article 36 of Protocol I to the Geneva Conventions requires countries to perform legal reviews of new weapons systems to determine whether their use would be prohibited under international law [14]. Israel is one of the few countries that have not ratified Protocol I [15]. UNESCO maintains that delegating life-or-death decisions to an AI system is itself forbidden [16]. Yet, no accountability framework exists to enforce this in the case of Habsora **TODO EVIDENCE NEEDED**.

Once deployed, establishing accountability becomes difficult. Under IHL, commanders are legally responsible for how weapons are used, even when autonomous functions are involved [17]. They are judged on what they knew at the time of the decision [18]. In practice, responsibility is spread across developers, the algorithm, the receiving IDF officer, and the approving commander [16]. The Lavender system, the human-targetting extension of Habsora, has a documented 10% error tolerance in targeting recommendations. XYZ This illustrates the problem: this margin makes civilian casualties inevitable. In the midst of war, real-time monitoring is not easy to pull off, which compounds the issue.

After deployment, we find that review processes are largely absent. The system’s opacity makes it difficult to determine whether casualties resulted from an AI recommendation or a human override. Palestinian civilians, who the most directly affected stakeholders, were never consulted. No structured ethical assessment, such as ‘value sensitive design’ (VSD), was conducted.

## 3 Responsible Solution Design 900

### 3.1 Lijst van problemen

#### 3.1.1 Technical

- Opaque ensemble, no XAI methods applied
- No reasoning chain
- Proportionality assessment impossible without understanding the basis of a recommendation
- Violations of P-01 (transparency) and P-03 (accountability)
- Compounded by CH-06 (lack of technical understanding of those who make the decisions) and CH-09 (absense of legal framework) (Khan et al. [10])

#### 3.1.2 Socio-technical

- Palestinians are indirect stakeholders who were never consulted → VSD failure
- Assymetry of consequencesbetween operators and Palestinians
- Commanders lack sufficient information and meaningful freedom of action → violation of two conditions of Meaningul Human Control (Steen et al. [7])
- 20-second review window makes critical evaluation impossible
- Automation bias, rubber-stamping

#### 3.1.3 Governance

- No Article 36 review because Israel has nto ratified Protocol I
- No ethical impact analysis done before deployment (for exampel VSD)
- Responsibility is fragmented, no clear accountability chain
- No real-time IHL compliance monitoring
- No post-deployment audit or incident review process
- Lavender extension has 10% error tolerance → civilian casualties inevitable

### 3.2 Mogelijke oplossing a.d.h.v. relevante literatuur

De 'nieuwe' literatuur komen uit het bestand 'RAI Assignment 2 Description' maar hebben we nog niet in de tekst verwerkt.

#### 3.2.1 Technical layer (XAI, interpretability)

- **Arrieta et al. [9]**: post-hoc solutions (SHAP, LIME, attention visualisation) applied to ensemble models
- **Miller [11]**:
- **(NEW) Liao, Gruen & Miller (2020)**: XAI question bank → operationalises user-centred explainability (Why this target? What evidence supports this recommendation?)
- **(NEW) Morley et al. (2020)**: maps ethics tools onto ML pipeline stages (data, model, deployment) so we can use this to say where and how speicifc XAI tools can be added before/during/after model development

#### 3.2.2 Socio-technical layer (MHC, automation bias, VSD, stakeholder inclusion)

- **Steen et al. [7]**: The 2 MHC conditions
- **Parasuraman and Manzey [13]**: VSD's 'tripartite' methodology (conceptual, empirical, technical) → concrete process for how stakeholders shold have been consulted (Palestinians were not consulted)
- **(NEW) Friedman, Kahn, Borning & Huldtgren (2013)**:
- **(NEW) Amershi et al. (2019)**: human-AI interaction guidelines (calibration of trust, commuinicating uncertainty, supporting error recovery) → design principles

#### 3.2.3 Governance (accountability, legal frameworks, organisational response)

- **Verdiesen et al. [8]**: addresses the empty cell in Figure 1 → required governance interventions (pre-deployment EIA, real-time IHL compliance, post-deployment audit)
- **(NEW) Stahl et al. (2022)**: Use the 3-level mitigation (policy-level strategies, guidance mechanisms, corporate governance) to categorise governance solutions
- **Wasilow and Thorpe [16]**: Canadian pre-deployment legal assessment protocols (Israel does not follow Article 36 review process)
- **Khan et al. [10]**: P-01/P-03/CH-06/CH-09 → we can use this to motiviate specific governance solutions

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